

End of Semester Wrap-Up

CMSC 104 Section 02
May 13, 2026

Administrative Notes

Classwork 8 is due tonight

Final is next Monday, May 20, 6-8 pm in this room

What did we cover this semester?

- “Computer Science”
 - What a computer is, and how to use it to solve problems
 - How to use the Python language to write programs to solve problems
 - This is not a Python boot-camp!!

Topics

- Hardware & software
- Machine, assembly & high-level languages
- System software vs application software
- Syntax vs Semantics
- Python details:
 - Variables
 - Operator precedence
 - If-elif-else
 - Loops: for each, for i, and while
 - Input and output
 - Types:
 - Int
 - String
 - Float
 - Boolean
 - List
 - Dictionary

Topics - Python (continued)

- Python
 - Operations on strings, lists and dictionaries
 - Functions
 - Mutable and immutable variable types
 - File I/O
 - Building larger programs from smaller units
 - How to develop software
 - Program design - simulation
 - Object-oriented programming
 - Python in an age of AI

Four cancelled classes!!

- Three due to weather, including the entire first week of class
- One due to instructor issue
- So what didn't we cover that we otherwise might have?
- Recursion
 - When a function calls itself
 - Major topic for computer science majors
 - Not so much for most people
- Algorithm analysis
 - You can write programs to solve problems
 - Some programs are “better” than others
 - They run faster
 - Take less memory
 - Don't require fast networks.

Topics we went light on...

In addition to skipping a couple of topics, we went lighter than normal on a couple

- Simulation: there's normally a classwork on the racquetball simulation work
- File reading & writing: we didn't get into cases where you need to read from a file; write to that or a different file; and then read from the new file

What you learned

But on the whole we covered almost everything we should have

We hope you have a good understanding of how to use your computer and the tools on it to solve problems that will occur in your academic and professional careers